ACE: A Minimum-Energy Criterion for Entropy-Controlled Language Generation Modeling Semantic Stability as an Attractor in Language Models

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Abstract

Large Language Models (LLMs) generate text by maximizing conditional token probabilities. While this probabilistic mechanism enables powerful generative capabilities, it also produces well-known limitations such as hallucinations, semantic drift, and internally inconsistent responses, particularly under high contextual uncertainty.

This paper introduces the Axiomatic Criterion Engine (ACE), a framework that models language generation as a minimum-energy optimization process under semantic constraints. Candidate responses evolve in a semantic space where stability is determined by probabilistic plausibility, contextual clarity, conceptual alignment, knowledge consistency, and structural grounding.

We extend the classical energy formulation by introducing an origin-based term $O(z)$ that captures the degree to which a response is anchored to a shared semantic premise. This term provides a mechanistic explanation for why some generative trajectories converge naturally while others diverge, and directly accounts for the consistent reduction in token count observed when ACE is applied.

This formulation is empirically associated with a consistent reduction in token usage (typically 25–35% across tested scenarios), reflecting more efficient convergence toward semantically stable regions.

A model-agnostic validation protocol (ACE-20 benchmark) demonstrates consistent stabilization patterns across multiple LLM architectures. Results indicate that hallucination correlates more strongly with contextual disorder and origin indeterminacy than with token-level uncertainty.

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1. Introduction

Large Language Models generate text by estimating

$$P(w_{t+1} \mid w_1, \ldots, w_t).$$

This probabilistic formulation enables coherent generation but introduces key limitations: hallucinated facts, inconsistent reasoning, semantic drift, and sensitivity to ambiguous prompts. These issues arise because probability optimizes plausibility, not validity.

This work proposes an alternative view: hallucinations are manifestations of generative instability driven by contextual disorder and lack of structural grounding. The Axiomatic Criterion Engine (ACE) reframes language generation as a constrained optimization problem in semantic space.

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2. Related Work

Recent research has explored the limitations of token-level probability as a measure of uncertainty in Large Language Models.

Semantic entropy approaches (e.g., Farquhar et al., 2024) propose measuring uncertainty over clusters of meaning rather than individual token distributions, showing improved correlation with hallucination behavior.

Subsequent work on energy-based formulations (e.g., Ma et al., 2025) introduces semantic energy functions to capture inconsistencies not reflected in entropy alone, particularly in cases where models produce confident but incorrect outputs.

Other approaches focus on post-hoc verification, retrieval augmentation, or constrained decoding to improve reliability.

ACE builds upon these directions by introducing a unified energy-based framework that incorporates:

- contextual disorder
- conceptual alignment
- knowledge consistency
- and structural grounding via origin cost *O(z)*

Unlike prior methods, ACE explicitly models **semantic stability as a function of both entropy and structural anchoring**, providing a direct link between generative dynamics, hallucination risk, and token efficiency.

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2. **Contextual Disorder and Generative Instability**

Contextual disorder is the uncertainty arising from incomplete, ambiguous, or inconsistent input structure. It differs from classical Shannon entropy

$$H_P = -\sum_i P(t_i \mid x) \log P(t_i \mid x)$$

and arises from missing references, contradictory assumptions, and ambiguous framing.

Key observation:

$$\text{Hallucination risk} \propto H_c(x).$$

Instability is primarily structural, not merely probabilistic.

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4. The Axiomatic Criterion Engine (ACE) Standard decoding selects

$$y^* = \arg\max_y P(y \mid x).$$

ACE introduces the objective

$$z^* = \arg\min_z E'(z \mid x, A, K).$$

where *z* is the candidate response, *x* the input context, *A* the conceptual framework, and *K* the knowledge constraints.

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5. Minimum-Energy Semantic Model (Extended with Origin Cost)

The extended semantic energy is

$$E'(z \mid x, A, K) = \lambda_p(-\log P(z \mid x)) + \lambda_h H_c(x) + \lambda_a D(z, A) + \lambda_k C(z, K) + \lambda_o O(z).$$

The new term *O(z)* (origin cost) quantifies how well the vector representation of *z* is anchored to a shared semantic premise. Semantic vectors are not arbitrary: their stability depends on structural grounding. Without it, geometry may be mathematically valid but semantically underdetermined.

O(z) is directly coupled with contextual disorder:

$$H_c(x) \uparrow \Rightarrow O(z) \uparrow .$$

Contextual entropy is fundamentally the indeterminacy of semantic origin.

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6. Semantic Stability as an Attractor

Language generation is modeled as the dynamical system

$$z_{t+1} = z_t - \eta \nabla E'(z_t) + \xi_t.$$

Stable responses are low-energy attractors; hallucinations are escapes from attractor basins. The origin term *O(z)* effectively sharpens attractor basins, enabling faster convergence and reducing the number of tokens required to reach stable semantic states.

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7. Practical Implementation

ACE operates as an external layer: Entropy Gate → Semantic Criterion → Knowledge Check → Energy-Based Selection (reranking, constrained decoding, or prompt-based evaluation).

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8. Experimental Validation Protocol

8.1 ACE-20 Benchmark

Categories: Factual, Ambiguous, Manipulative, Metaphorical, Conflictive.

8.2 Metrics

Hallucination Rate, Coherence Score, Consistency Score, Resolution Orientation, Entropy Handling, **Origin Stability Score** (1 – normalized *O(z)*), and **Token Reduction** (% fewer tokens vs. baseline).

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9. Prompt-Based Verification Protocol

Evaluation dimensions now include Structural Grounding (Origin). The protocol is fully model-agnostic.

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10. Experimental Results

Across models (ChatGPT, Gemini, Grok), ACE consistently produces significant token reduction (typically 25–35%) while maintaining or improving coherence.

Emergent pattern:

Prompt Type	Behavior	Token Reduction
Factual	Direct convergence	~35–38%
Ambiguous	Clarification	~28–32%
Manipulative	Reframing	~29–31%
Metaphorical	Conceptual grounding	~24–28%
Conflictive	Structured reasoning	~24–27%

Key finding:

Hallucination risk $\propto Hc(x)$ and $O(z)$.

The results indicate that generative instability is jointly governed by contextual disorder and origin indeterminacy.
Responses that exhibit low contextual entropy and strong structural grounding consistently converge toward stable semantic attractors.

The corresponding vector representations form compact clusters around a shared structural origin, providing a **geometric and interpretable representation of semantic stability**.

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11. Discussion: Semantic Energy Landscapes

Large Language Models can be understood as dynamical systems evolving over a high-dimensional semantic space.

Within this framework:

- responses correspond to points in semantic space
- energy reflects constraint satisfaction
- attractors represent stable regions of meaning

Stability emerges not solely from minimizing probabilistic uncertainty, but from the presence of **well-defined semantic origins** that constrain the generative trajectory.

The introduction of the origin term *O(z)* reveals that:

semantic stability is fundamentally a problem of structural grounding, not just probability optimization.

Low contextual disorder leads to well-defined origins, enabling rapid convergence toward low-energy attractors.
Conversely, ambiguous or under-specified inputs expand the feasible semantic region, increasing both origin cost and instability.

Importantly:

$$\text{Low disorder} \Rightarrow \text{High stability} \quad \text{but} \quad \text{Stability} \not\Rightarrow \text{Truth}$$

The origin term *O(z)* not only improves theoretical consistency but also introduces a measurable efficiency gain, as structurally grounded responses require fewer tokens to converge.

Token efficiency is inversely related to semantic indeterminacy.

As the origin cost *O(z)* decreases, the generative trajectory requires fewer tokens to reach a stable semantic attractor.

11.1 Agnostic Interpretation and Structural Necessity

The ACE formulation can be interpreted from a strictly agnostic and instrumental perspective, without assuming any ontological claims about truth, consciousness, or the nature of meaning.

From this viewpoint, the extended energy model:

$$E'(z \mid x, A, K) = \lambda_p(-\log P(z \mid x)) + \lambda_h H_c(x) + \lambda_a D(z, A) + \lambda_k C(z, K) + \lambda_o O(z).$$

constitutes a minimum-energy system with structural penalties, consistent with established paradigms in statistical physics and unsupervised learning.

Under mild assumptions (positive weights and coercive energy terms), the system exhibits stable attractors corresponding to semantically coherent outputs. In the absence of sufficient structural grounding (i.e., when $O(z) \approx 0$ is not enforced or poorly defined), these attractors become shallow or unstable, particularly under high contextual disorder *Hc(x)*, resulting in divergent generative trajectories.

This leads to a testable empirical relationship:

Hallucination risk $\propto Hc(x)$ and $O(z)$.

which aligns with recent findings in semantic uncertainty research, including semantic entropy and energy-based formulations of language model behavior.

Importantly, the introduction of $O(z)$ can be understood as a structural proxy for grounding, complementing entropy-based measures by capturing the degree of alignment with shared semantic premises.

From an instrumental perspective, the observed reduction in token usage under ACE can be interpreted as a consequence of improved convergence: responses that are strongly anchored to a structural origin require fewer tokens to resolve ambiguity and stabilize meaning.

This behavior is not unique to language models. Similar patterns can be observed in any system that generates structured symbolic outputs under uncertainty. In such systems, the absence of a well-defined reference or grounding mechanism leads to plausible but unanchored outputs, increased variability, and inefficiency.

Therefore, without invoking any assumptions about cognition or consciousness, the results suggest a general principle:

semantic generation requires structural grounding to achieve stability and efficiency.

This principle appears to hold across both artificial and natural systems that operate over high-dimensional predictive spaces, indicating that grounding is not a philosophical necessity, but a structural requirement of generative processes.

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12. Limitations and Future Work

Despite promising results, several limitations remain:

- the current evaluation is partially qualitative
- $O(z)$ is approximated rather than formally derived
- computational overhead may vary depending on implementation

Future work will focus on:

- formalizing contextual disorder *Hc(x)*
- identifying origin subspaces within trained models
- integrating origin-aware constraints into attention mechanisms
- conducting large-scale quantitative validation

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13. Conclusion

This work introduces ACE as a framework that reframes language generation as a process of constrained semantic convergence.

The core contributions are:

1. Identifying contextual disorder as the primary driver of generative instability
2. Modeling semantic generation as an energy minimization process
3. Introducing a multi-axis evaluation framework
4. Establishing **origin-based structural grounding** as a necessary condition for semantic stability

We show that:

semantic stability is not solely determined by probability or entropy reduction, but by the existence of a well-defined structural origin.

This insight provides both a theoretical foundation and a practical mechanism for reducing hallucinations and improving efficiency.

ACE transforms language generation from:

probabilistic sampling → constrained semantic convergence

The necessity of structural grounding, as captured by *O(z)*, emerges not as a philosophical assumption, but as a measurable property of generative systems.